

ASSIGNMENT #02

Course: CHE 301: CHEMICAL ENGINEERING THERMODYNAMICS
Term: Fall 2025
Professor: Dr. Michael Caracotsios
Office: EIB 244
Cell Phone: (630) 696-0934
E-mail: mcaracot@uic.edu

Names : Iliana Perez and Queen Adedire

Due Date: Wednesday, September 17th in Class (11:00AM)

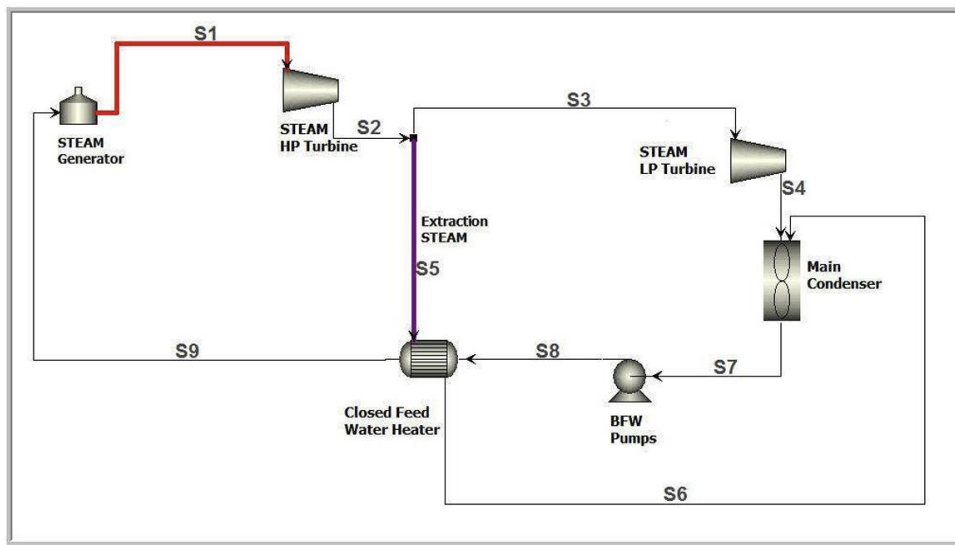
INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
- Include this cover page with your name(s) in the report.
- Upload your report as a PDF file by the Due Date and Time on the UIC Blackboard.
- Bring a hard copy of your report as a PDF file by the Due Date and Time in class

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

Assumptions/Problem Set Up

Problem A: Superheated steam is generated in a boiler at $P = 80 \text{ bar}$ and $T = 480^\circ \text{C}$ and expanded into a High-Pressure Turbine. A slip stream (known as extraction steam) is diverted into a Closed Feed Water Heater (CFWH) to heat the liquid condensate (Stream S8) before it enters the steam generator. The cold extraction steam (Stream S6) is then directed back to the main condenser. The rest of the HP Turbine exhaust is expanded into a steam condensing turbine (CST) to a pressure $P = 8 \text{ kPa}$. This process is known as **Regenerative Rankine Cycle** with Closed Feed Water Heater (CFWH) and it is very common in modern power plants because it results in substantial reduction of fuel usage and therefore carbon dioxide emissions. A schematic of a typical **Regenerative Rankine Cycle** with CFWH is shown below:



The following operating data are available for this power plant:

1. Total power generated from the two turbines: 1000 MW
2. HP turbine discharge pressure: 15 bar
3. Temperature of Stream S7 (subcooled liquid) 38 deg C
4. HP and LP Turbine efficiency: 80%
5. Boiling Feed Water (BFW) pump efficiency 70%
6. Fuel used: Bituminous Coal with heating value 22 MJ/kg
7. Boiler efficiency: 80%
8. Temperature of Steam S9 is 5°C lower than that of stream S6 which is saturated liquid.

To set up this problem correctly first we need to make some assumptions about the nine different streams in the cycle. The problem tells us that S1 has a pressure, $P_1 = 80 \text{ bar}$, and a temperature $T_1 = 480^\circ \text{C}$. We are not given a flow rate for the steam generated in the boiler, S1, this is something we will be solving for, but to start out and set up the problem we can assume that $m_1 = 1 \frac{\text{kg}}{\text{s}}$. We also are told that the thermodynamic phase of S1 is super-heated vapor.

Moving onto Stream 2, S2, this stream is the discharge of the HP turbine. The problem tells us that for this stream $P_2 = 15 \text{ bar}$. This stream is the direct output of S1 and no other streams are added so it is safe to assume that the molar flow rate is the same and so based off our initial assumption $m_2 = 1 \frac{\text{kmol}}{\text{s}}$.

S2 is diverted into S3 and S5, nothing is present that would change the pressures of each stream, so we assume that $P_2 = P_3 = P_5 = 15 \text{ bar}$ also $T_2 = T_3 = T_5$. The divergence of S2 into S3 and S5 does change the molar flow rates. We made the initial guess that based off $m_2 = 1 \frac{\text{kmol}}{\text{s}}$ that the molar flow rate was split into $m_3 = 0.8 \frac{\text{kmol}}{\text{s}}$ and $m_5 = 0.2 \frac{\text{kmol}}{\text{s}}$. These values are just the guesses we chose because we do know that $m_3 + m_5 = m_2$.

S4 is the discharge of the LP turbine, the problem statement tells us that the LP turbine expands the steam resulting in a pressure of $P_4 = 8 \text{ kPa} = 0.08 \text{ bar}$. Since S4 is the output of S3 and no other flow rates are entering the turbine, we can assume that the molar flow rate stays the same meaning $m_3 = m_4 = 0.8 \frac{\text{kmol}}{\text{s}}$. We do not know temperature of S4 but in the textbook, in the Section "Example 2.2: The Rankine Cycle" it states that the quality of steam outside of a Condensing Steam Turbine (CST) is $x=0.9$. The LP turbine is a CST meaning that we can assume the quality of S4 is also 0.9.

Now we look at S6, which is the output of S5 into the Closed Feed Water Heater (CFWH). In the CFWH streams don't mix which means that the pressure and molar flow rate of S6 are equal to its input S5. We know can assume that $m_6 = m_5 = 0.2 \frac{\text{kmol}}{\text{s}}$ and $P_6 = P_5 = 15 \text{ bar}$. The problem statement tells us that S6 is a saturated liquid, this information can be used to find the temperature of S6 using the WaterTSat function in the CHE calculator.

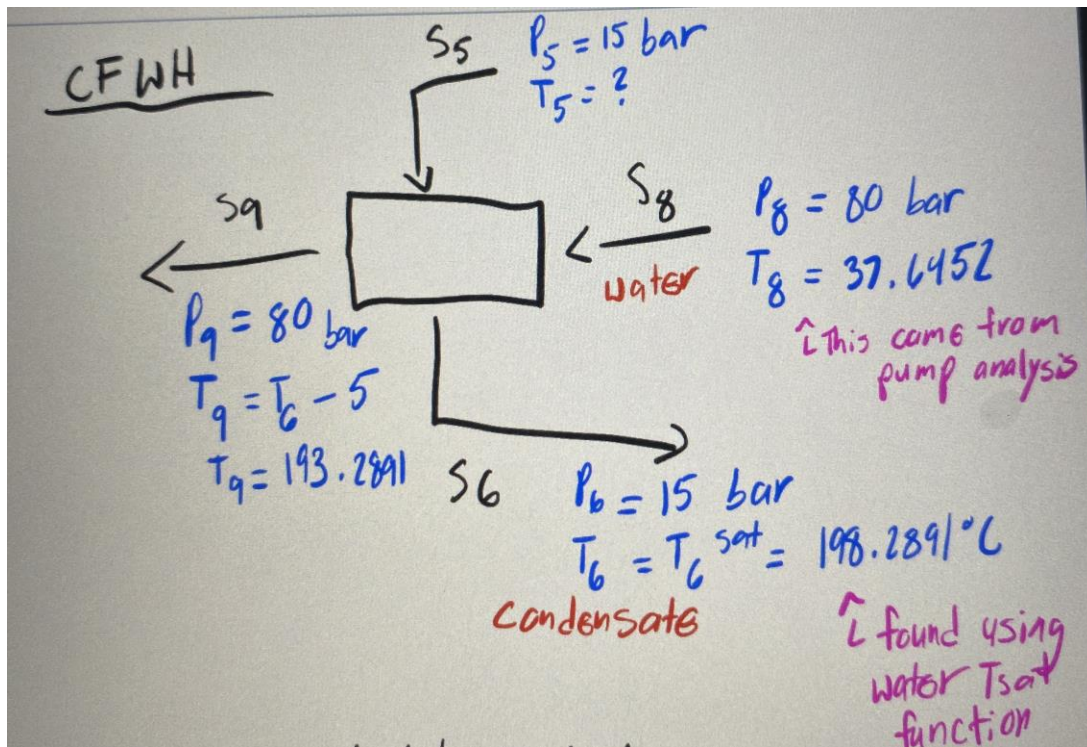
Both S6 and S4 enter into the main condenser and come out as S7. To find the mass flow rate of S7 we need to add $m_6 + m_4 = 0.2 + 0.8 = 1 \frac{\text{kmol}}{\text{s}}$. For the pressure as we discussed in lecture, when two pressures are combined the resulting stream will take the minimum pressure value, $P_6 = 15 \text{ bar}$ and $P_4 = 0.08 \text{ bar}$, the resulting pressure of S7 will be $P_7 = 0.08 \text{ bar}$. The problem statement tells us that S7 is a subcooled liquid with a temperature of $T_7 = 38 \text{ C}$.

Now looking at S8 it is the outcoming stream from S7 through the pump, and the pump does work that raises the pressure, so we know the pressure of S8 must rise from the pressure of S7

which was 0.08 bar. The molar flow rate of S7 through the pump does not change so $m_7 = m_8 = 1 \frac{\text{kmol}}{\text{s}}$. We also know that the pump only deals with liquid, and the phase does not change meaning S8 is also a sub cooled liquid. To assume the pressure, we look at the outlet stream through the CFWH, which is S9. S9 feeds into the steam generator which produces S1 and there is nothing that would affect the pressure from S9 to S1, so it is safe to assume that $P_1 = P_9 = 80 \text{ bar}$. Working backwards the streams in the CFWH do not mix so the pressure of S9 and S8 must also remain the same, it safe to further assume that $P_1 = P_9 = P_8 = 80 \text{ bar}$. The problem also tells us that $T_9 = T_6 - 5$, using the waterTsat function to find the temperature of S6 we can then find the temperature of S9.

Now, using all these assumptions we can begin to formulate the calculations in the CHE calculator for all nine streams. For the outlet streams of the turbines and pumps we will have to calculate conditions both under the isentropic and real conditions, to correctly calculate the work.

Closed Feed Water Heater Analysis



Cooling water is available at $P_8 = 80 \text{ bar}$ and $T_8 = 37.65^\circ\text{C}$ is being heated with steam available at $P_5 = 15 \text{ bar}$ and $T_5 = ?$ (this is the temperature we are using the energy balance of the CFWH to find).

Let's assume that the steam flow rate is $m_5 = 0.2 \frac{\text{kmol}}{\text{s}}$ and that of the cold water is $m_8 = 1 \frac{\text{kmol}}{\text{s}}$, this is based on our assumptions from the problem set up. When the steam exits the heat exchanger it is saturated liquid, and it is known as condensate. The temperature of the condensate is the saturated steam temperature at $P_6 = 15 \text{ bar}$. This temperature can be calculated using the WaterTsat function:

$$T_6^{sat} = \text{WaterTsat}(P_6) \rightarrow T_6^{sat} = 198.2891 \text{ } ^\circ\text{C}$$

Now let's write the total energy balance using the hot steam and the cold water as the combined system: $-\Delta mH + \dot{Q} + W_s = 0$

For the combined system there is no external heat load going in or out and also there is no mechanical work done on it. Therefore, the total energy balance is simplified to:

$$-\Delta mH = 0 \rightarrow m_5 H_5 + m_8 H_8 - m_6 H_6 - m_9 H_9 = 0$$

From the mass balance on the hot steam and on the cold water separately we obtain

$$m_5 = m_6 \text{ and } m_8 = m_9$$

Since the mass flow rates are known we have one equation (the energy balance) with one unknown cold water outlet temperature. Let's define a procedure on how to calculate the unknown temperature from the total energy balance:

1. In the first step, since we know the conditions of the cooling water stream, we can calculate the enthalpy and the entropy at the inlet and outlet conditions. Using the CHE Calculator we obtain:

Cold Water into Heat Exchanger: S8

$$H_8 = H\text{Steam}(T_8, P_8) = 2967.425 \frac{\text{kJ}}{\text{kmol}}$$

$$S_8 = S\text{Steam}(T_8, P_8) = 9.685 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}$$

Cold Water out of Heat Exchanger: S9

$$H_9 = H\text{Steam}(T_9, P_9) = 14866.524 \frac{\text{kJ}}{\text{kmol}}$$

$$S_9 = S\text{Steam} = (T_9, P_9) = 40.662 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}$$

2. In the next step we must guess the steam inlet temperature say $T_5 = 40^\circ\text{C}$ and then calculate the conditions of the condensate stream both in the inlet and the outlet of the heat exchanger. Using again the CHE Calculator we obtain:

Steam into the Heat Exchanger: S5

$$H_5 = H_{\text{steam}}(T_5, P_5,) = \text{initial guess}$$

$$S_5 = S_{\text{steam}}(T_5, P_5,) = \text{initial guess}$$

Steam Condensate out of Heat Exchanger: S6

$$H_6 = H_{\text{steam}}(, P_6, \text{liquid}) = 15216.608 \frac{\text{kJ}}{\text{kmol}}$$

$$S_6 = S_{\text{steam}}(, P_6, \text{liquid}) = 41.695 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}$$

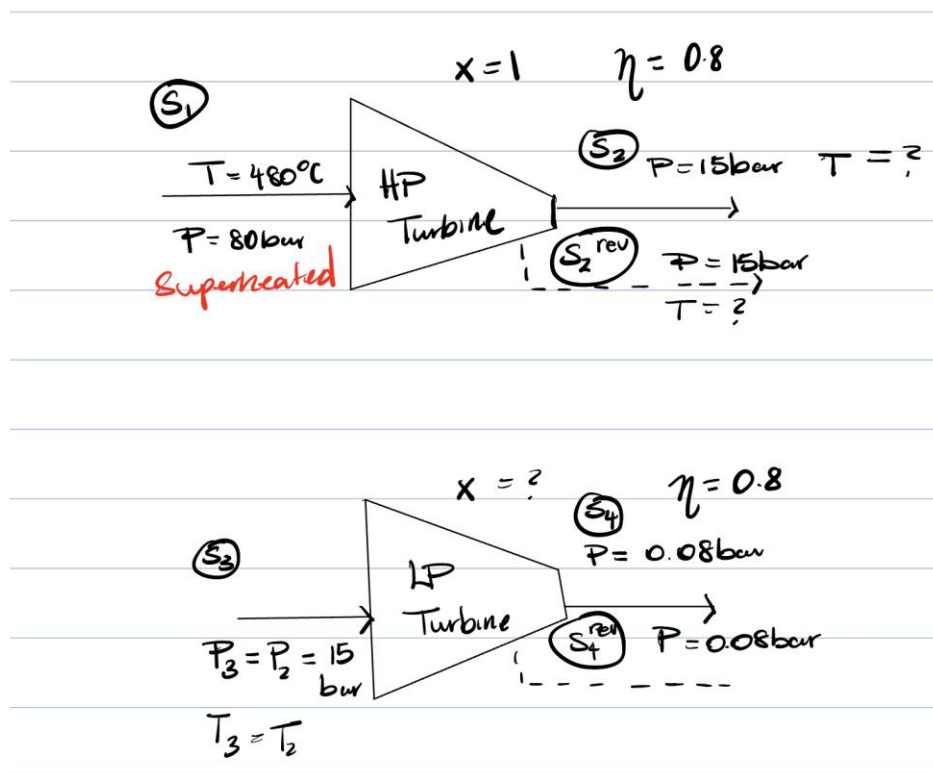
3. In the next step we formulate the total energy balance

$$-\Delta mH = m_5 H_5 + m_8 H_8 - m_6 H_6 - m_9 H_9 = 0$$

4. In the final step we GoalSeek the energy balance to zero by manipulating the steam inlet temperature and obtain $T_5 = 785.6301^\circ\text{C}$

*** Note that these values were done on the original basis of 1 kmol/s once we solved the the turbine and the actual amount of steam generated, they changed and gave us the actual temp for T5, which can be seen in our CHE calculations worksheet.

Turbine Analysis



HP Turbine

For Stream 1 since we are given the temperature of 480°C and a pressure of 80 bar, we can calculate the molar enthalpy and molar entropy using Hsteam and Ssteam functions respectively.

$$H_1 = 60342.81 \frac{\text{kJ}}{\text{kmol}}$$

$$S_1 = 120.011 \frac{\text{kJ}}{\text{kmol}}$$

For stream 2 (rev) we aren't given the temperature, but we are given pressure of 15 bar. We guess a temperature, say 40°C then calculate the enthalpy and entropies using this temperature. We need to set $S_1 = S_2^{\text{rev}}$ by using an entropy balance equation.

$$S_1 - S_2^{\text{rev}} = 0$$

We Goal Seel by setting the entropy equation equal to 0 by changing the temperature.

$$T_2^{\text{rev}} = 239.7^\circ\text{C}$$

$$W_2^{\text{rev}} = -511811.59 \text{ kJ}$$

For the actual stream 2 we also guess an initial temperature, find the molar enthalpy and entropy using this value and find the work

$$\dot{W}_2 = -409449.29 \text{ kW}$$

The thermal efficiency is then calculated using this equation

$$\eta = \frac{\dot{W}}{\dot{W}^{rev}}$$

Since this value is not equal to the value given 0.8, we can come up with another equation $\eta - 0.8 = 0$.

Use goal seek to set this value equal to zero by changing temperature we get a new temperature of $T_2 = 277.82^\circ\text{C}$

LP Turbine

The inlet stream 3 = stream 2 for temperature and pressure, which make the enthalpies and entropies equal

$$H_2 = H_3 = 53842.56 \frac{\text{kJ}}{\text{kmol}}$$

$$S_2 = S_3 = 123.07 \frac{\text{kJ}}{\text{kmol}}$$

$$\text{Enthalpy flow} = m_3 \cdot H_3$$

$$\text{Entropy flow} = m_3 \cdot S_3$$

The outlet stream 4

We are given that the pressure is 0.08 bar and no temperature to find the enthalpy and entropy we need to have a reversible stream for stream 4.

We initially guess a vapor fraction for stream 4 to be 0.7, calculate the enthalpy of the vapor and liquid phase then use the lever rule to find the molar enthalpy, the same can be done for the entropy.

$$H_4^{rev} = H_{\text{steam}}(0.08, v)$$

$$H_4^{rev} = H_{\text{steam}}(0.08, l)$$

$$\text{Lever Rule: } H_4^{rev} = xH_v + (1 - x)H_l$$

We then use Goal-Seek to set

$$S_3 = S_4^{rev}$$

By changing the quality.

In the actual stream 4 we want the efficiency to be 0.8 as given so we again

Guess the quality, calculate the entropies and enthalpies like stream 4 rev, and find the work of both the reversible and the actual stream. We then use the efficiency

equation $\eta = \frac{\dot{W}_4^{rev}}{\dot{W}_4}$ Goal-Seek $\eta - 0.8 = 0$ by changing the efficiency of stream 4.

Since we made assumptions for all the molar flowrates and their split fractions, we need to find the split fraction equation for stream 5 and 3 using the energy balance of the CFWH rearranging this set of equations:

$$m_5 \cdot H_5 + m_8 \cdot H_8 - m_6 \cdot H_6 - m_9 \cdot H_9 = 0$$

$$m_2 = m_5 + m_3$$

We get the split fraction equations to be

$$\frac{m_5}{m_2} = \frac{H_9 - H_8}{H_5 - H_6} = 0.305$$

$$\frac{m_3}{m_2} = 1 - \frac{m_5}{m_2} = 0.695$$

To find the actual molar flowrates for all the streams

We use:

$$\frac{|W_{LP\ Turbine} + W_{HP\ Turbine}|}{1000} - 1000$$

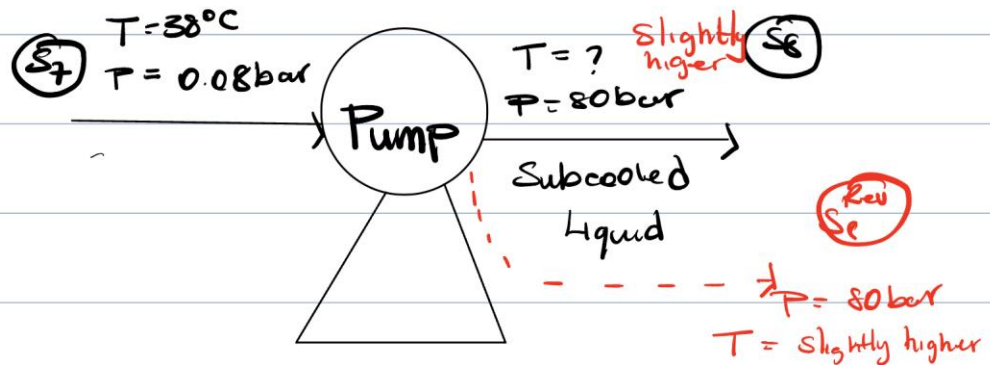
Goal Seek this to equal zero by changing the molar flowrate of stream 1 we get

$$m_1 = 66.55 \frac{kmol}{s}$$

$$m_5 = 20.32 \frac{kmol}{s}$$

$$m_3 = 46.23 \frac{kmol}{s}$$

Pump Analysis



We are given that stream 7 has a temperature of 38°C and from our assumptions stream 7 has a pressure of 0.08 bar and stream 8 has a pressure of 80 bar, the temperature is unknown. We need to calculate (a) the temperature of the steam at the discharge (S_8) of the pump (b) the power requirement by the pump and (c) the lost work, assuming that the steam flow rate is $m_7 = m_8 = 66.55\text{ kmol/s}$. To accomplish that we will start with the macroscopic energy balance

$$-\Delta(mH) + \dot{Q} + W_s = 0$$

Assuming that the heat loss (gain) is small compared to the enthalpy of the steam, then $\dot{Q} = 0$. Therefore, we call the process adiabatic, and the macroscopic energy balance becomes:

$$-\Delta(mH) + \dot{W} = 0$$

We notice that we have only one equation but two unknowns: the discharge temperature and the power required by the pump. Therefore, we need one more equation. That will be the macroscopic entropy balance

$$-\Delta(mS) + \sum \frac{\dot{Q}}{T} + S_{gen} = 0$$

Since the process is adiabatic the entropy balance becomes:

$$-\Delta(mS) + S_{gen} = 0$$

If we assume that the pump operation is reversible then $S_{gen} = 0$ and therefore the macroscopic entropy balance further simplifies to: $-\Delta(mS) = 0$

Since we have steady conditions, $-\Delta m = 0 \rightarrow m_7 = m_8 = m = \frac{1 \text{ kmol}}{s}$ and therefore by using the lever rule the entropy balance becomes:

$$-\Delta m(\Delta S) = 0 \rightarrow S_7 = S_8^{rev}$$

where the superscript rev indicates that the pump operation is assumed reversible and therefore the entropy generation is equal to zero. For the reversible process we can calculate the power required by the pump by using the macroscopic energy balance:

$$-\Delta m(\Delta H) + W_s^{rev} = 0 \rightarrow W_s^{rev} = m(H_8^{rev} - H_7)$$

Stream 7 entropy and enthalpy can be calculated using the given pressure and temperature.

Now that we have the formulas ready, let's see how we can calculate the W_s^{rev} . For this calculation we need the temperature of the discharge stream S8. Since we do not know the value, we must guess one.

Knowing that the temperature rise across the pump tends to be very small, we assume a value of $T_8^{rev,(0)} = 40^\circ\text{C}$. We now calculate the enthalpy and entropy of stream S8 at this value.

Since the entropy of the stream S8 for the reversible pump operation is not the same as the entropy of the stream S7, we need to guess another value for the stream S8. Instead of guessing new values we use Goal Seek to solve the equation by manipulating the temperature T_8^{rev} . Doing that we find that $T_8^{rev} = 41.78^\circ\text{C}$ and the corresponding values of molar enthalpy and entropy:

$$H_8^{rev} = 3,010.85 \frac{\text{kJ}}{\text{kmol}}$$

$$S_8^{rev} = 9.824 \frac{\text{kJ}}{\text{kmol}}$$

Now we can calculate the reversible work from:

$$W_8^{rev} = m(H_8^{rev} - H_7) \rightarrow W_8^{rev} = 9632.51 \text{ kW}$$

The actual power that is required by the pump is:

$$\dot{W}_8 = m(H_8 - H_7) \rightarrow \dot{W}_8 = 13760.73 \text{ kW}$$

Since the actual power for the process is different from the ideal, the enthalpy of stream S8 is different than that of the calculated value for the reversible process, which means that the stream S8 temperature is different (greater) than the temperature calculated for the reversible process is $T_8^{rev} = 38.22$ To calculate the new temperature and the power requirement, we follow the steps outline below:

1. We first guess a new temperature of stream S8 say, $T_8^{(0)} = 50^\circ\text{C}$
2. Next, we calculate the enthalpy and entropy of stream S8: H_8, S_8
3. Next, we calculate the power required by the pump: $\dot{W}_s = m(H_8 - H_7)$
4. Next, we formulate the efficiency equation: $\eta\dot{W}_s - \dot{W}_s^{rev} = 0$
5. Finally, we use GoalSeek to make the efficiency equation equal to zero, by manipulating the temperature of the stream S8

Performing the tasks indicated in the steps above we obtain: $T_8 = 41.2^\circ\text{C}$ At this temperature we have:

$$H_8 = 3072.88 \frac{\text{kJ}}{\text{kmol}}$$

$$S_8 = 10.023 \frac{\text{kJ}}{\text{kmol}}$$

To calculate the lost work, first we calculate the availability of the inlet and outlet streams using the equations:

$$B_7 = H_7 - T_0 S_7 \text{ and } B_8 = H_8 - T_0 S_8$$

where $T_0 = 300\text{K}$ and then using the lost work balance equation

$$LW = -\Delta m(H - T_0 S) + \dot{W}_s = m(B_7 - B_8) + \dot{W}_s$$

to obtain $LW = 3972.13 \text{ kW}$

Calculations:

1. The amount of steam required in MMSCFD

This calculation was done through the analysis of the turbines and pumps and the work is discussed there. The value is 4553.09 MMSCD. This value was found using the conversion function in the CHE Calculator KMOLperSECToMMSCFD to convert $\dot{m}_1 = 66.55 \frac{\text{kmol}}{\text{s}}$ to the correct units 4810 MMSCD.

2. The amount of Bituminous coal required in kg/s

In this part of the calculations, we are trying to figure out the mass flow rate of coal that is needed to be supplied to the system. To solve the mass flow rate, we need to calculate the amount of heat being added to the fluid in the boiler which can be called $\dot{Q}_{in}$, this is just the heat from the combustion of the coal. The equation for calculating $\dot{Q}_{in}$ is:

$$\dot{Q}_{in} = m_{steam} \times \Delta H_{boiler}$$

The m_{steam} is the amount of steam required that we previously solved for, which is $66.55 \frac{kmol}{s}$. The change in enthalpy of the boiler is $H_{steam} - H_{feedwater} = H_1 - H_9$.

$$\dot{Q}_{in} = 66.55 \frac{kmol}{s} \times \left(60342.08 - 14866.524 \frac{kJ}{kmol} \right) = 3026445.74 \frac{kJ}{s}$$

Now, we can use the heat being added to the fluid to find the total energy that must be supplied to the coal $\dot{Q}_{fuel}$, which is based on the heat added to the fluid and the boiler efficiency, which are both known:

$$\dot{Q}_{fuel} = \frac{\dot{Q}_{in}}{\eta_{boiler}} = \frac{\left(3026445.74 \frac{kJ}{s} \right)}{0.8} = 3783057.18 \frac{kJ}{s}$$

Now that we know the amount of heat that must be supplied to the coal, we can use the given heating value of the coal (22 MJ/kg) to find the mass flow rate of the coal. Note that we used the conversion factor of 1000 kJ = 1 MJ to convert $\dot{Q}_{fuel}$ from kJ/s to MJ/s to have the correct units with the LHV.

$$m_{coal} = \frac{\dot{Q}_{fuel}}{LHV} = \frac{3783 \frac{MJ}{s}}{22 \frac{MJ}{kg}} = 172 \frac{kg}{s}$$

The amount of coal required is 172 kg/s

**** The derivations for the equations needed to calculate the amount of coal came from ChatGPT*****

3. The lost work in each equipment of the Regenerative Rankine Cycle

Component	Lost Work (kW)
Boiler	1497508.4881
HP Turbine	61045.2866
LP Turbine	135233.21
Condenser	300311.3531
Pump	3972.1334
CFWH	115508.1311

The equations for the Lost Work of all the components are based on the general equation for lost work:

5.) Lost work.
 Balance, $-\Delta m(H - T_0 S) + \sum Q_i \left(1 - \frac{T_0}{T_i}\right) + W_s = 0$
 Availability
 $T_0 = \text{Temp of infinite environment}$
 $\rightarrow 300 \text{ K}$
 A calculation that work is always lost in universe

The lost work for each component is derived from this general balance.

LW for Boiler:

Starting by looking at the boiler, there is no shaft work but there is heat so the equation for Lost Work (LW) will simplify to:

$$LW = \dot{Q}_H \left(1 - \frac{T_0}{T_H}\right) + \dot{W}_{S, \text{TURBINE}} + \dot{W}_{S, \text{PUMP}}$$

We know our $\dot{Q}_H$ value because that is the $\dot{Q}_{in}$ value we previously solved for. It is assumed that T_0 is always 300 K and T_H is the temperature required to ignite coals into flames to heat the boiler, according to ChatGPT we can assume the value $T_H = 1400^\circ\text{C} = 1673 \text{ K}$. Using the values we collected from our calculations in the CHE calculator we plugged this equation in to get the value 1497508.4881

kW.

LW for Turbines:

For the turbines there is no heat but there is shaft work so the general equation for LW is:

$$B_1 = H_1 - T_0 S_1 \quad \text{and} \quad B_2 = H_2 - T_0 S_2$$

where $T_0 = 300 \text{ K}$ and then using the lost work balance equation

$$LW = -\Delta m(H - T_0 S) + \dot{W}_s = m(B_1 - B_2) + \dot{W}_s$$

Where B is the availability which can be calculated using enthalpies and entropies as shown below. We calculated the availability and the shaft work in the CHE calculators so we just plugged all those values that can be seen in our ENG calculations worksheet into the equation for LW in excel and got the LW for LP turbine as 135233.21 kW and the HP turbine as 61045.2866 kW.

LW for Condenser:

In the condenser there is no work being done but there is heat and so the LW equation simplifies to:

$$LW = -\Delta m(H - T_0 S) + \dot{Q}_C \left(1 - \frac{T_0}{T_C}\right) = m[(H_2 - T_0 S_2) - (H_3 - T_0 S_3)]$$

Using the availabilities and the mass flow rate that went into the condenser (S4 and S6) and the exhaust stream coming out (stream 7) we used the values in the CHE calculator and plugged them into this equation and got a LW value of 300311.3531 kW.

LW for Pump:

In the pump there is no heat but there is work being done therefore the equation for the LW is:

To calculate the lost work, first we calculate the availability of the inlet and outlet streams using the equations:

$$B_3 = H_3 - T_0 S_3 \quad \text{and} \quad B_4 = H_4 - T_0 S_4$$

where $T_0 = 300 \text{ K}$ and then using the lost work balance equation

$$LW = -\Delta m(H - T_0 S) + \dot{W}_S = m(B_3 - B_4) + \dot{W}_S$$

The corresponding streams for the pump in our problem were stream 8 going into the pump and stream 9 coming out of the pump. Using those availabilities and the work done by the pump calculated in the CHE calculator we plugged this equation into the excel calculations and got a LW value of 3972.1334 kW.

LW for CFWH:

The CFWH has no heat or work being done on this system, so the LW Equation simply is:

$$LW = -\Delta m(H - T_0 S) = -\Delta m B$$

Where B is the availability. We know from the mass balance on the CFWH that $m_5 = m_6$ and $m_8 = m_9$. This equation is stating that the LW equals mass times availability in-out so into the calculator we plugged in $LW = m_5(B_5 - B_6) + m_8(B_8 - B_9) = 115508.1311$

kW.

4. The thermal and thermodynamic efficiency of the cycle if the goal is power generation

To find the thermal efficiency we use the equation:

$$\eta_{thermal} = \frac{|W_{total}|}{Q_{in}}$$

The total work is the work done by all the components $W_{turbines} + W_{pump} = -432590.39 + -567409.61 + 13760.73 \frac{kJ}{kmol}$

Now we can solve for the thermal efficiency

$$\eta_{thermal} = \frac{-432590.39 + -567409.61 + 13760.73 \frac{kJ}{kmol}}{2864548.267 \frac{kJ}{s}} = 0.33$$

The thermal efficiency is 0.33

** This equation came from the textbook

To find the thermodynamic efficiency we use the equation:

$$\eta_{thermodynamic} = \frac{W_{net}}{W_{rev}}$$

This based off the net work done on the pumps and turbines divided by the work that would be done in a reversible ideal condition. These numbers for work all came from the calculations we did in the CHE calculator for work based on the enthalpies of the streams going into the turbines and pumps.

$$\eta_{thermodynamic} = \frac{-432590.39 + -567409.61 + 13760.73 \frac{kJ}{kmol}}{-540737.97 + -709262 + 9632.51 \frac{kJ}{kmol}} = 0.90$$

This equation was derived from ChatGPT**

The thermodynamic efficiency is 0.90.

5. The emissions of CO₂ in gigatons per year.

****All the assumptions and equations in this problem are from ChatGPT****

$$m_{coal} = 172 \frac{kg}{s}$$

We can assume that coal is 70-80% carbon by mass leading to a conversion factor. Now we know that if we react Carbon and oxygen gas it will produce CO₂ emissions, $C + O_2 \rightarrow CO_2$, using the balanced equation we can now use the molar masses of carbon and carbon dioxide and divide

them by each other $\frac{44 \frac{kg}{kmol} CO_2}{12 \frac{kg}{kmol} C} = 3.667 \text{ kg of } CO_2 \text{ per kg } C$

Using all this information, we solve for carbon emissions:

$$m_{CO_2} = 172 \frac{kg}{s} \cdot 0.75 \cdot 3.667 = 473 \frac{kg}{s}$$

Now we need to convert to the correct units of gigatons per year:

$$473 \frac{kg}{s} \times 31,536,000 \frac{s}{year} \times \frac{1 \text{ GT}}{10^{12} kg} = 0.015 \frac{GT}{yr}$$

The CO₂ emissions are 0.015 GT/yr

6. Find through research the CO₂ emissions per year worldwide in gigatons. Make sure you report the year for which you have found this information

By researching online, we found on the website “Statista” that the worldwide CO₂ emissions were 37.41 gigatons per year in 2024.

Material and Energy Balances

Stream Properties	stream 1	Stream 2(rev)	stream 2	stream 3	Stream 4 (rev)	stream 4	Stream 5	stream 6	stream 7	ream 8(ides	stream 8	steam 9
Molar Flow, kmol/s	66.5500	66.5500	66.5500	46.2303	46.2303	46.2303	20.3197	20.3197	66.5500	66.5500	66.5500	66.5500
Mass Flow, kg/s												
Volume Flow, m³/s												
Temperature, C	480.0000	239.7419	277.8231	277.8231			277.8231	198.2891	38.0000	38.2248	39.0527	193.2891
Pressure, bar	80.0000	15.0000	15.0000	15.0000	0.0800	0.0800	15.0000	15.0000	0.0800	80.0000	80.0000	80.0000
Vapor Fraction	1.0000	1.0000	1.0000	1.0000	0.8169	0.8878	1.0000	0.0000				
Molecular Weight, kg/kmol												
Molar Enthalpy, kJ/kmol	60342.808	52217.517	53842.575	53842.575	38500.649	41569.034	53842.575	15216.608	2866.106	3010.847	3072.879	14866.524
Vapor Phase	60342.808	52217.517	53842.575	53842.575	46426.672	46426.672	53842.575	0.000	46426.672	0.000	0.000	
Liquid Phase	0.000	0.000	0.000	0.000	3132.177	3132.177	0.000	15216.608	3132.177	3010.847	3072.879	
Molar Entropy, kJ/(kmol K)	120.011	120.011	123.068	123.068	123.068	132.819	123.068	41.695	9.824	9.824	10.023	40.662
Vapor Phase	120.011	120.011	123.068	123.068	148.256	148.256	123.068	0.000	148.256	0.000	0.000	
Liquid Phase	0.000	0.000	0.000	0.000	10.675	10.675	0.000	41.695	10.675	9.824	10.023	
Molar Density, kmol/m³												
Work												
Enthalpy Flow, kJ/s	4015812.60	3475074.63	3583222.21	2489158.96	1779896.95	1921749.35	1094063.25	309196.43	190739.31	200371.82	204500.04	989366.87
Entropy Flow, kJ/(s K)	7986.71	7986.71	8190.20	5689.49	5689.49	6140.27	2500.71	847.23	653.82	653.82	667.06	2706.08
Availability, kJ/s	24339.58	16214.29	16922.06	16922.06	1580.13	1723.31	16922.06	2708.03	-81.23	63.51	65.86	2667.81
Duty [Power], kJ/s [kW]		-540737.97	-432590.39		-709262.01	-567409.61				9632.51	13760.73	
Efficiency			0.80			0.80					0.70	
Compressibility Factor												
Qin	3026445.7358											

User Data and Calculations	
Lost Work Boiler (kW)	1497508.4881
Lost work of Turbine 1 (kW)	61045.2866
Lost Work Turbine 2 (kW)	135233.2100
Lost Work Condenser (kW)	300311.3531
Lost Work Pump (kW)	3972.1334
Lost Work CFWH (kW)	115508.1311
Energy Balance CFWH	0.0000
Entropy Balance Turbine	0.0000
Efficiency equation turbine	0.0000
Split Fraction M5	0.3053
Split Fraction M3	0.6947
Power Equation	0.0000
Entropy Balance Pump	0.0000
Efficiency Equation Pump	0.0000
Entropy Balance Turbine 2	0.0000
Efficiency Balance Turbine 2	0.0000
amount of steam, MMSCFD	4810.4155
Thermal Efficiency	0.3259
Thermodynamic Efficiency	0.8978

